

# Notes for Complex Analysis

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## Abstract

This is a set of notes and comments on the classic book *Complex Analysis* by Kunihiro Kodaira [1]. I tried to apply Kodaira's method described in his autobiography:

"In my view, mathematics books (including research papers) are the most obscure and difficult texts to read. Reading a math book of several hundred pages from cover to cover is an incredibly daunting task. The moment you open one, you are instantly bombarded with definitions, axioms, theorems, and proofs.

Mathematics is the kind of discipline where everything becomes crystal clear once you understand it. Therefore, when I read a math book, I usually look only at the theorems, try my best to understand their statements, and then attempt to figure out the proofs on my own. Most of the time, however, I find myself completely baffled. I am eventually forced to look at the author's proof. Yet, there are times when even after reading the proof repeatedly, its meaning remains elusive. In such situations, I will try copying the mathematical proofs by hand into my notebook.

During the process of copying, I often notice parts of the proof that seem unsatisfactory, which prompts me to think about whether alternative methods of proof exist. By the time I finally manage to finish a chapter this way, I have completely forgotten everything that came before it. Having no other choice, I simply have to start reviewing all over again from the beginning."

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# Preliminaries

## 0.1 Linear transformation

Which one of the following is true?

1. If  $f$  is a  $\mathbb{R}$ -linear map from  $\mathbb{C}$  to  $\mathbb{C}$ , then  $f$  is also a  $\mathbb{C}$ -linear map.
2. If  $f$  is a  $\mathbb{C}$ -linear map from  $\mathbb{C}$  to  $\mathbb{C}$ , then  $f$  is also a  $\mathbb{R}$ -linear map.

The second is TRUE, and the first is FALSE. In fact,  $\mathbb{C}$ -linear is much stronger than  $\mathbb{R}$ -linear. We look at the definitions. Let  $f : \mathbb{C} \rightarrow \mathbb{C}$ , and take arbitrary  $z, w \in \mathbb{C}$ . Then

1.  $f$  is  $\mathbb{R}$ -linear iff for any  $a, b \in \mathbb{R}$ ,

$$f(az + bw) = af(z) + bf(w)$$

2.  $f$  is  $\mathbb{C}$ -linear iff for any  $\alpha, \beta \in \mathbb{C}$ ,

$$f(\alpha z + \beta w) = \alpha f(z) + \beta f(w)$$

So clearly, if 2 holds, then we may restrict  $\alpha, \beta$  to be real numbers and 1 holds.

**Example 0.1.1.**  $f(z) := \bar{z}$  is NOT  $\mathbb{C}$ -linear. To be  $\mathbb{C}$ -linear,  $f$  has to satisfy  $f(iz) = if(z)$ . But  $\overline{iz} = \overline{i(u + iv)} = u\bar{i} - \bar{v} = -iu + v = -i(u - iv) = -i\bar{z}$ .

Intuitively, this is because, if you regard  $\mathbb{C}$  as  $\mathbb{R} \times \mathbb{R}$ , i.e., a two-dimensional  $\mathbb{R}$ -vector space, then a  $\mathbb{R}$ -linear map can be an arbitrary  $2 \times 2$  matrix. However, if you regard  $\mathbb{C}$  as a one-dimensional  $\mathbb{C}$ -vector space, then a  $\mathbb{C}$ -linear map  $f$  can only be the form of  $f(z) = f(1)z$ .

However, we may ask, under what circumstance, a  $\mathbb{R}$ -linear map  $f : \mathbb{C} \rightarrow \mathbb{C}$  is also a  $\mathbb{C}$ -linear map? Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ :

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix}$$

To make  $f$  a  $\mathbb{C}$ -linear map,  $f$  suffices to satisfy:

$$f(iz) = if(z)$$

where  $z = x + iy$ . Now, what is the matrix  $\mathbf{J}$  representing left multiplication by  $i$ ? Note that

$$i(a + ib) = -b + ia$$

and

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -b \\ a \end{pmatrix}$$

So  $\mathbf{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . Therefore,  $\mathbf{A}$  suffices to satisfy  $\mathbf{AJ} = \mathbf{JA}$ . That is:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

This implies that  $\mathbf{A}$  has to be of the form:

$$\begin{pmatrix} p & q \\ -q & p \end{pmatrix}$$

Now, let  $f : \mathbb{C} \rightarrow \mathbb{C}$  be a map. Write  $f(x, y) = u(x, y) + iv(x, y)$ . Recall that  $f$  is  $\mathbb{R}$ -differentiable iff the Jacobian

$$\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$$

is some real-valued matrix. But if we require  $f$  to be a  $\mathbb{C}$ -differentiable function, then the local linear approximation is not only  $\mathbb{R}$ -linear, but necessarily  $\mathbb{C}$ -linear. That is, we deduce the Cauchy-Riemann equations.

# 1 Holomorphic Functions

## 1.1 Holomorphic functions

### 1.1.1 The complex plane

We omit this sub-subsection.

### 1.1.2 Functions of a complex variable

We omit this sub-subsection.

### 1.1.3 Holomorphic functions

Let  $D \subseteq \mathbb{C}$  be a region and  $f$  a function from  $D$  to  $\mathbb{C}$ . Let  $z \in D$  and let  $\rho$  be a positive real number s.t. the  $\rho$ -neighborhood

$$U_\rho(z) := \{\xi : |\xi - z| < \rho\}$$

is contained in  $D$ . If

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists, the function  $f$  is said to be *differentiable* at  $z$ . This limit is called the *differential coefficient* of  $f(z)$ , denoted by  $f'(z)$ . If  $f(z)$  is differentiable at all points  $z \in D$ , then  $f'(z)$  is called the derivative of  $f(z)$ . Now, note that:

$$f(z+h) - f(z) = f'(z)h + O(h)$$

Therefore, if  $f(z)$  is differentiable, then

$$\lim_{h \rightarrow 0} f(z+h) = f(z) \text{ for every } z \in D$$

In other words, differentiable functions are continuous.

Putting

$$w = f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

and

$$h := \Delta x + i\Delta y, \quad \Delta w := f(z+h) - f(z) = \Delta u + i\Delta v, \quad f'(z) := p + iq$$

Then

$$\Delta u + i\Delta v = (p+iq)(\Delta x + i\Delta y) + O(h)$$

Splitting this into real and imaginary parts, we have<sup>1</sup>

$$\begin{aligned} \Delta u &= u(x+\Delta x, y+\Delta y) - u(x,y) = p\Delta x - q\Delta y + O(\dots) \\ \Delta v &= v(x+\Delta x, y+\Delta y) - v(x,y) = q\Delta x + p\Delta y + O(\dots) \end{aligned}$$

So if  $f(z) = u(x,y) + iv(x,y)$  is a differentiable function of  $z$ , then  $u(x,y)$  and  $v(x,y)$  as functions of the real variable  $x, y$  are differentiable and

$$u_x(x,y) = v_y(x,y) = p \quad \text{and} \quad u_y(x,y) = -v_x(x,y) = -q$$

Hence

$$u_x(x,y) = v_y(x,y) \quad u_y(x,y) = -v_x(x,y) \tag{1}$$

The equations 1 are called the **Cauchy-Riemann equations**. As a direct consequence, we note

$$f' = p + iq = u_x + iv_x = v_y - u_y$$

Conversely, let  $f(z) = u(x,y) + iv(x,y)$  be a function of  $z = x + iy \in D \subseteq \mathbb{C}$ , s.t. its real part  $u(x,y)$  and imaginary part  $v(x,y)$  are differentiable satisfying the relations 1. Then, putting

$$p := u_x(x,y) = v_y(x,y) \quad \text{and} \quad q := -u_y(x,y) = v_x(x,y)$$

we have

$$\begin{aligned} \Delta u &= u_x(x,y)\Delta x + u_y(x,y)\Delta y + O(\dots) \\ &= p\Delta x - q\Delta y + O(\dots) \\ \Delta v &= v_x(x,y)\Delta x + v_y(x,y)\Delta y + O(\dots) \\ &= q\Delta x + p\Delta y + O(\dots) \end{aligned}$$

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<sup>1</sup>... here is  $|h| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$

Evaluating  $f(z+h) - f(z)$  for  $h = \Delta x + i\Delta y$  we get

$$f(z+h) - f(z) = \Delta u + i\Delta v = (p+iq)(\Delta + i\Delta y) + O(\dots)$$

Hence

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = p + iq$$

That is,  $f(z)$  is differentiable and  $f'(z) = p + iq$ . Summing up, the function  $f = u + iv$  is differentiable of  $z = x + iy$  if and only if  $u, v$  are differentiable of  $x, y$ , satisfying the Cauchy-Riemann equations 1.

**Definition 1.1.1.** Let  $f(z)$  be a function of the complex variable  $z$  defined on  $D \subseteq \mathbb{C}$ . If  $f$  is a differentiable function of  $z$  s.t.  $f'$  is continuous, then  $f(z)$  is called a **holomorphic function** of  $z$ .

*Remark.* Usually, a function is called *holomorphic* if it is differentiable. Later we will show that the derivative of a differentiable function of a complex variable is always continuous, so that our definition is equivalent with the usual one.

The following theorem follows directly:

**Theorem 1.1.2.** A function  $f : D \subseteq \mathbb{C} \rightarrow \mathbb{C}$  written in our usual form<sup>2</sup> is holomorphic if and only if its real part  $u(x, y)$  and its imaginary part  $v(x, y)$  are continuously differentiable over the real  $x$  and  $y$ , satisfying the C-R equations 1.

**Example 1.1.3.**  $z$  is obviously a holomorphic function over  $z$ . However,  $\bar{z}$  is NOT. For putting  $h = \Delta x + i\Delta y$ , we have

$$r := \frac{\overline{z+h} - \bar{z}}{h} = \frac{\bar{h}}{h} = \frac{\Delta x + i\Delta y}{\Delta x - i\Delta y}$$

Note that if  $\Delta y = 0$ , then  $\lim_{h \rightarrow 0} r = 1$ . But if  $\Delta x = 0$ , then  $\lim_{h \rightarrow 0} r = -1$ . So  $\lim_{h \rightarrow 0} r$  does not exist.

**Example 1.1.4.** Let  $f(z) = \frac{z}{|z|^2}$ . i.e.,

$$f(x + iy) = \frac{x}{x^2 + y^2} + i \frac{y}{x^2 + y^2}$$

Is  $f$  holomorphic?

**Theorem 1.1.5.** If the complex functions  $f(z), g(z)$ , defined on a domain, are holomorphic, then their linear combination  $a_1 f(z) + a_2 g(z)$ , and the product  $f(z)g(z)$  are also holomorphic. If, moreover,  $g(z) \neq 0$  on the domain, then the quotient  $f(z)/g(z)$  is holomorphic, too.

It follows from the above example and theorem that, the polynomial  $p(z)$  of  $z$

$$p(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n, \quad a_0, \dots, a_n \in \mathbb{C}$$

is holomorphic on  $\mathbb{C}$ . Moreover, the rational expression

$$\frac{p(z)}{q(z)} \quad \text{when } p(z), q(z) \text{ are polynomials}$$

is holomorphic on the domain, obtained by omitting from the complex plane  $\mathbb{C}$  the roots of the algebraic equation  $q(z) = 0$ .

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<sup>2</sup> $f(z) = u(x, y) + iv(x, y)$  where  $z = x + iy$

## 1.2 Power series

### 1.2.1 Series whose terms are functions

Let  $\{f_n(z)\}$  be a sequence of functions all defined on the same domain. Let  $D$  be the subset of that domain. If this sequence converges for each point of  $D$ , then we say that  $\{f_n(z)\}$  converges on  $D$ . In that case, a function

$$f(z) := \lim_{n \rightarrow \infty} f_n(z)$$

is defined on  $D$ .

**Definition 1.2.1.** If for any  $\epsilon > 0$ , there exists  $n_0(\epsilon) \in \mathbb{N}$  s.t.

$$|f(z) - f_n(z)| < \epsilon \quad \text{whenever } n > n_0(\epsilon)$$

for all  $z \in D$ , then the sequence of functions  $\{f_n(z)\}$  is said to *converge uniformly* to  $f(z)$ .

Similarly, if the series  $\sum_{n=1}^{\infty} f_n(z)$  converges for each  $z \in D$ , then the series  $\sum_{n=1}^{\infty} f_n(z)$  is said to converge on  $D$ .

If the function sequence  $\{s_m(z)\}$  of partial sums  $s_m(z) := \sum_{n=1}^m f_n(z)$  is uniformly convergent on  $D$ , then the series  $\sum_{n=1}^{\infty} f_n(z)$  is said to be uniformly convergent on  $D$ . Further, if the series  $\sum_{n=1}^{\infty} |f_n(z)|$  is convergent on  $D$ , then the series  $\sum_{n=1}^{\infty} f_n(z)$  is said to be absolutely convergent on  $D$ . (In this case, obviously  $\sum_{n=1}^{\infty} f_n(z)$  is convergent on  $D$ .) If the series  $\sum_{n=1}^{\infty} |f_n(z)|$  converges uniformly on  $D$ , then the series  $\sum_{n=1}^{\infty} f_n(z)$  is said to be uniformly absolutely convergent. From

$$|s_m(z) - s_n(z)| = \left| \sum_{n=m+1}^{\infty} f_n(z) \right| \leq \sum_{n=m+1}^{\infty} |f_n(z)|$$

we know that if  $\sum_{n=1}^{\infty} f_n(z)$  is uniformly absolutely convergent on  $D$ , then  $\sum_{n=1}^{\infty} f_n(z)$  is uniformly convergent on  $D$ .

**Theorem 1.2.2.** Let  $f_1(z), \dots, f_n(z), \dots$  be continuous functions on  $D$ . Then

- (i). If the sequence  $\{f_n(z)\}$  converges uniformly on  $D$ , then its limit  $f(z) = \lim_{n \rightarrow \infty} f_n(z)$  is continuous on  $D$ .
- (ii). If the series  $\sum_{n=1}^{\infty} f_n(z)$  converges uniformly on  $D$ , then its sum  $\sum_{n=1}^{\infty} f_n(z)$  is continuous on  $D$ .

*Proof.* Suppose (i) holds. Then, if the series  $\sum_{n=1}^{\infty} f_n(z)$  converges uniformly on  $D$ , we have the partial sums  $\{s_n(z)\}$  converges uniformly on  $D$ . Hence by (i),

$$\sum_{n=1}^{\infty} f_n(z) = \lim_{n \rightarrow \infty} s_n(z)$$

is continuous on  $D$ . Therefore, it suffices to prove (i). Let  $c$  be an arbitrary point of  $D$  and let us prove that  $f(z)$  is continuous at  $c$ . Fix an arbitrary  $\epsilon > 0$ , then there exists  $n \in \mathbb{N}$  s.t.

$$|f_n(z) - f(z)| < \epsilon \quad \text{for all } z \in D$$

Since  $f_n(z)$  is continuous at  $c$ , there exists  $\delta(\epsilon) > 0$  s.t.

$$|f_n(z) - f_n(c)| < \epsilon \quad \text{whenever } |z - c| < \delta(\epsilon)$$

Note that

$$|f(z) - f(c)| < |f(z) - f_n(z)| + |f_n(z) - f_n(c)| + |f_n(c) - f(c)| < 3\epsilon$$

whenever  $|z - c| < \delta(\epsilon)$ , as desired.  $\square$

### 1.2.2 Power series

Power series play a fundamental role in complex analysis. In this section, we consider power series

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots a_n z^n + \cdots$$

with center at the origin 0. By replacing  $z$  by  $z - c$ , all results obtained will be seen to be valid for power series centered at  $c$  as well. We want to find a sufficient condition for the power series  $\sum_{n=0}^{\infty} a_n z^n$  to converge.

Suppose that  $\sum_{n=0}^{\infty} a_n z^n$  converges for a certain complex number  $z$ . In this case  $\lim_{n \rightarrow \infty} a_n z^n = 0$ , hence there exists a  $M > 0$  s.t.

$$|a_n z^n| \leq M \quad \text{for all } n \in \mathbb{N}$$

For such an  $M$  we have

$$|z| |a_n|^{1/n} \leq M^{1/n}$$

while  $\lim_{n \rightarrow \infty} M^{1/n} = 1$  (so  $\limsup_{n \rightarrow \infty} M^{1/n} = 1$ .) Therefore taking the  $\limsup_{n \rightarrow \infty}$  on both sides we have

$$|z| \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq 1$$

If  $0 < \limsup_{n \rightarrow \infty} |a_n|^{1/n} < +\infty$ , let

$$r := \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}}$$

and we obtain

$$|z| \leq r$$

If  $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = +\infty$ , then let  $r := 0$ . The number  $r$  defined in this way has the following property:

**Theorem 1.2.3.** *The power series  $\sum_{n=0}^{\infty} a_n z^n$  converges absolutely if  $|z| < r$  and diverges if  $|z| > r$ .*

*Proof.* If  $|z| < r$ , then we choose a real number  $\kappa$  s.t.  $|z| < \kappa < r$ . Then

$$\frac{1}{\kappa} > \frac{1}{r} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$$

Therefore, there exists  $n_0 \in \mathbb{N}$  s.t.  $|a_n|^{1/n} < 1/\kappa$  whenever  $n > n_0$ , i.e.  $|a_n|\kappa^n < 1$  for  $n > n_0$ . Therefore, there exists a constant real  $M$  s.t.

$$|a_n| \leq \frac{M}{\kappa^n}$$

Since  $|z| < \kappa$ , we have

$$\sum_{n=0}^{\infty} |a_n z^n| = \sum_{n=0}^{\infty} |a_n| |z^n| \leq \sum_{n=0}^{\infty} \frac{M}{\kappa^n} |z^n| = M \sum_{n=0}^{\infty} \left( \frac{|z|}{\kappa} \right)^n = \frac{M\kappa}{\kappa - |z|} < \infty$$

as desired. The second part is immediate.  $\square$

The real number  $r$  is called the **radius of convergence (r.o.c.)** of the power series  $\sum_{n=0}^{\infty} a_n z^n$ . Putting  $1/+\infty = 0$  and  $1/0 = +\infty$ , the r.o.c. is given in all cases by

$$r = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}} \quad (2)$$

(2) is called the **Cauchy-Hadamard formula**. For  $0 < r < +\infty$  the circle  $C_r$  with center 0 and radius  $r$  is called the **circle of convergence** of the power series  $\sum_{n=0}^{\infty} a_n z^n$ . According to the above theorem, the series converges absolutely if...; and diverges if...

If  $z$  is a point *on*  $C_r$ , then  $\sum_{n=0}^{\infty} a_n z^n$  can be convergent or divergent. The interior of  $C_r$  is denoted by

$$U_r(0) := \{z : |z| < r\}$$

If  $r = +\infty$ , the circle of... is not defined, but we put  $U_{+\infty}(0) = \mathbb{C}$  and consider  $\mathbb{C}$  as the interior of the circle of... for all radii of convergence s.t.  $0 < r \leq +\infty$ .

**Theorem 1.2.4.** *Let  $\sum_{n=0}^{\infty} a_n z^n$  be a power series with r.o.c.  $r$  s.t.  $0 < r \leq +\infty$ . If  $\rho$  is a real number s.t.  $0 < \rho < r$ , then  $\sum_{n=0}^{\infty} a_n z^n$  is uniformly absolutely convergent on the interior  $U_\rho(0)$  of the circle 0 and radius  $\rho$ .*

*Proof.* Choose a real  $\kappa$  s.t.  $\rho < \kappa < r$ . Note that  $|z| < \rho < r$ , by our previous discussion, for sufficiently large  $n$  we have:

$$|a_n| \rho^n \leq \frac{M}{\kappa^n} \rho^n = M(\rho/\kappa)^n$$

for some constant  $M$ . Since

$$\sum_{n=0}^{\infty} M(\rho/\kappa)^n = M\kappa/(\kappa - \rho) < +\infty$$

we have

$$\sum_{n=0}^{\infty} |a_n| \rho^n < +\infty$$

Note that  $|a_n z^n| \leq |a_n| \rho^n$  for all  $z$  with  $|z| < \rho$ , the statement follows.  $\square$

**Theorem 1.2.5.** Let  $\sum_{n=0}^{\infty} a_n z^n$  be a power series with r.o.c.  $r$  s.t.  $0 < r \leq +\infty$ . Then  $f(z) = \sum_{n=0}^{\infty} a_n z^n$  is a holomorphic function of  $z$  on the interior  $U_r(0)$  and the derivative of  $f(z)$  is given by termwise differentiation:

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

Moreover, the r.o.c. of the power series  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  is also  $r$ .

*Proof.* We first show that the r.o.c. of the series  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  equals  $r$ . Differentiating both sides of the following equality

$$\sum_{n=0}^m t^n = \frac{1 - t^{m+1}}{1 - t}$$

yields

$$\sum_{n=1}^m n t^{n-1} = \frac{1 - (m+1)t^m + m t^{m+1}}{(1-t)^2}$$

Since  $\lim_{m \rightarrow \infty} m t^m = 0$  for  $0 < t < 1$ , we have

$$\sum_{n=1}^m n t^{n-1} = \frac{1}{(1-t)^2}$$

Now, in order to prove that  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  converges absolutely if  $|z| < r$ , we fix a  $z$  with  $|z| < r$ . Next we choose a real  $\kappa$  s.t.  $|z| < \kappa < r$ . We know that there exists a constant  $M$  s.t.

$$|a_n| \leq M/\kappa^n$$

Hence

$$\sum_{n=1}^{\infty} n |a_n z^{n-1}| \leq \frac{M}{\kappa} \sum_{n=1}^{\infty} n \left( \frac{|z|}{\kappa} \right)^{n-1} = \frac{M\kappa}{(\kappa - |z|)^2} < +\infty$$

Therefore,  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  converges absolutely for  $|z| < r$ . From

$$\sum_{n=0}^{\infty} |a_n z^n| \leq |a_0| + |z| \sum_{n=1}^{\infty} n |a_n z^{n-1}|$$

we conclude that  $\sum_{n=0}^{\infty} a_n z^n$  converges absolutely if  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  converges absolutely. Therefore, the r.o.c. of  $\sum_{n=1}^{\infty} n a_n z^{n-1}$  equals  $r$ .

Next, we show that  $f(z) = \sum_{n=0}^{\infty} a_n z^n$  is differentiable on  $U_r(0)$  and that  $f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$ . See Page 20 of the book...  $\square$

So we conclude that  $f'(z)$  is holomorphic of  $z$  on  $U_r(0)$  and

$$f''(z) = \sum_{n=2}^{\infty} n(n-1) a_n z^{n-2}$$

Since the r.o.c. of this power series again equals  $r$ ,  $f''(z)$  is also holomorphic on  $U_r(0)$  and

$$f'''(z) = \sum_{n=3}^{\infty} n(n-1)(n-2)a_n z^{n-3}$$

Continuing in this way, we find that  $f(z)$  is infinitely often differentiable on  $U_r(0)$ ... Replacing  $z$  by  $z - c$  and applying the above results to the power series  $\sum_{n=0}^{\infty} a_n(z - c)^n$ , we get the following result:

Let  $\sum_{n=0}^{\infty} a_n(z - c)^n$  be a power series with r.o.c.  $r$ , such that  $0 < r \leq +\infty$ . Then  $\sum_{n=0}^{\infty} a_n(z - c)^n$  is absolutely convergent for  $|z - c| < r$  and divergent for  $|z - c| > r$ . The r.o.c. is given by the Cauchy-Hadamard formula (2). If  $0 < r < \infty$ . Consequently,

**Theorem 1.2.6.** *The sum of the power series*

$$f(z) = \sum_{n=0}^{\infty} a_n(z - c)^n = a_0 + a_1(z - c) + a_2(z - c)^2 + \dots$$

*is a holomorphic function of  $z$  on the interior  $U_r(c)$  of the circle of convergence. The function  $f(z)$  is infinitely differentiable on  $U_r(c)$  and the derivatives  $f'(z), f''(z), \dots$ , are all holomorphic on  $U_r(c)$ . The  $m^{\text{th}}$  derivative of  $f(z)$  is*

$$f^{(m)}(z) = \sum_{n=m}^{\infty} n(n-1)(n-2)\dots(n-m+1)a_n(z - c)^{n-m}$$

*and the r.o.c. of this power series is also  $r$ .*

Substituting  $z = c$  in the above expression, we have

$$f^{(m)}(c) = m!a_m$$

Therefore, the original power series can be written as

$$f(z) = \sum_{n=0}^{\infty} a_n(z - c)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (z - c)^n$$

The power series on the RHS is called the **Taylor series** of  $f(z)$  with center  $c$ .

Now, let  $f(x)$  be a real analytic function of  $x$ , defined on some interval of the real line  $\mathbb{R}$ . Let  $c$  be a point from the interior of that interval and let  $f(x)$  be represented by its Taylor series with center  $c$ :

$$f(x) = f(c) + \sum_{n=1}^{\infty} \frac{f^{(n)}(c)}{n!} (x - c)^n$$

Assuming that its r.o.c.  $r$  satisfies  $0 < r \leq +\infty$ , then the power series obtained from this Taylor series by replacing  $x$  by the complex variable  $z$  converges absolutely in the interior  $U_r(c)$  in the complex plane  $\mathbb{C}$  and represents a holomorphic function of  $z$  on  $U_r(c)$ . Denoting this function also by  $f$ , we have

$$f(z) = f(c) + \sum_{n=1}^{\infty} \frac{f^{(n)}(c)}{n!} (z - c)^n$$

The original real analytic function is obtained by replacing the complex variable  $z$  by the real variable  $x$ , i.e., by restricting  $f(z)$  to the real axis. A real analytic function can always be extended to a holomorphic function.

**Example 1.2.7.** The exponential function  $e^x$  has a Taylor expansion

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

with r.o.c. equal to  $+\infty$ . Therefore, the exponential function of the complex variable  $z$

$$e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots$$

is holomorphic on the whole complex plane  $\mathbb{C}$ .

The complex sine function

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} \pm \cdots$$

and the complex cosine function

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} \pm \cdots$$

are also holomorphic on the whole complex plane  $\mathbb{C}$ .

Since  $\log(1+x)$  has the Taylor expansion

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

with r.o.c. 1, the complex extension

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} \pm \cdots$$

is holomorphic on the interior of the unit circle  $U_1(0)$ .

We note as an obvious consequence of the above, the following formula

$$e^{iz} = \cos z + i \sin z$$

In particular for real  $\theta$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

so that  $|e^{i\theta}| = 1$ . Hence the transformation given by

$$z \rightarrow z' = e^{i\theta} z$$

is a rotation of the complex plane  $\mathbb{C}$  with center 0 and angle  $\theta$ . Every complex number  $z \neq 0$  can be represented as

$$z = |z|e^{i\theta}$$

where  $\theta$  is the angle ...

## 1.3 Integrals

In the previous section we have seen that the sum

$$f(z) = \sum_{n=0}^{\infty} a_n(z - c)^n$$

is a holomorphic function of  $z$  in the interior of the circle of convergence. Conversely, it is possible to represent a holomorphic function, defined on a certain region, in some neighborhood of an arbitrary point  $c$  of that region as the sum of a power series, i.e.,

$$f(z) = \sum_{n=0}^{\infty} a_n(z - c)^n \quad \text{if } |z - c| < r \text{ for some } r > 0$$

### 1.3.1 Curves

Consider a complex-valued function  $\gamma(t)$ , defined on some interval of the real line  $\mathbb{R}$ . The following definitions are obvious. If for each  $t$  in the interior of the interval

$$\lim_{\Delta t \rightarrow 0} |\gamma(t + \Delta t) - \gamma(t)| = 0$$

then  $\gamma(t)$  is called a *continuous function* of  $t$ . If for each  $t$  in the interior of the interval, the limit

$$\gamma'(t) = \lim_{\Delta t \rightarrow 0} \frac{\gamma(t + \Delta t) - \gamma(t)}{\Delta t}$$

exists, then  $\gamma(t)$  is called a *differentiable function* of  $t$ .

If  $\gamma(t)$  is differentiable and its derivative  $\gamma'(t)$  is continuous, then  $\gamma(t)$  is called *continuously differentiable* or a *function of class  $\mathbf{C}^1$* . The function  $\gamma(t)$  is continuous, of class  $\mathbf{C}^1$  and so on iff its real part  $\Re \gamma(t)$  and its imaginary part  $\Im \gamma(t)$  are both continuous, of class  $\mathbf{C}^1$ , and so on.

**Definition 1.3.1.** A continuous map  $\gamma : t \mapsto \gamma(t)$  from the closed interval  $I = [a, b]$  into  $\mathbb{C}$  is called an **arc** or **curve** in the complex plane.

If  $\alpha$  and  $\beta$  are two points of  $\mathbb{C}$ , then a curve  $\gamma : t \mapsto \gamma(t), t \in [a, b]$  s.t.  $\gamma(a) = \alpha, \gamma(b) = \beta$  is said to *connect*  $\alpha$  and  $\beta$ . If  $D$  is a region in  $\mathbb{C}$ ,  $[D]$  its closure,  $\alpha \in D$  and  $\beta \notin [D]$ , then all curves connecting  $\alpha$  and  $\beta$  intersect the boundary  $[D] - D$ . That is, for some  $s \in (a, b)$  we have  $\gamma(s) \in [D] - D$ . To see this, let  $S$  be the subset of  $I$  consisting of all points  $t$  s.t.  $\gamma(t) \notin [D]$  and let  $s$  be the supremum of  $S$ . Suppose  $\gamma(s) \notin [D]$ . Then by the continuity of  $\gamma$  there exists an  $\epsilon > 0$  s.t.  $\gamma(t) \notin [D]$  whenever  $t > s - \epsilon$ . This is a contradiction and therefore  $\gamma(s) \in [D]$ . Suppose  $\gamma(s) \notin D$ , then there exists an  $\epsilon > 0$  s.t.  $\gamma(t) \notin D$  for  $t < s + \epsilon$ . This is a contradiction and therefore  $\gamma(t) \in D$ .

Hence, if it is possible to connect the points  $\alpha$  and  $\beta$  of  $\mathbb{C}$  with a curve  $\gamma$  not intersecting the boundary of  $D$ , then either  $\alpha$  and  $\beta$  belong to  $D$  or to the complement of  $[D]$ .

If  $\gamma : t \mapsto \gamma(t), t \in I = [a, b]$  is a curve s.t.  $\gamma$  is injective from  $I$  to  $\mathbb{C}$ , then  $\gamma$  is called a **Jordan arc**. Topologically, a Jordan arc is a curve that homeomorphic to  $[0, 1]$ . (it has distinct endpoints and never crosses itself).

If  $\gamma$  is such that  $\gamma(a) = \gamma(b)$ , then  $\gamma$  is called a **closed curve**. That is, it forms a loop, but it is allowed to self-intersect (e.g., a figure-8 or a tangled loop is still a closed curve, as long as the ends connect).

If  $\gamma$  is such that for all  $s, t$  with  $a \leq s < t \leq b$  we have  $\gamma(s) \neq \gamma(t)$ , while  $\gamma(a) = \gamma(b)$ , then  $\gamma$  is called a **Jordan curve**. Topologically, a Jordan curve is a *simple* (never self-intersects) and *closed* (ends connect) curve that is homeomorphic to a circle.

*Remark.* Note that there are examples of curves, which are not "curve-like" at all, such as a triangle, together with all of its interior points in  $\mathbb{C}$ .

Jordan arcs of class  $\mathbf{C}^1$ , however, have properties that are worthy of curves. Let  $\gamma : t \mapsto \gamma(t), t \in [a, b]$ , be a Jordan arc of class  $\mathbf{C}^1$ , and let  $t_0 \in [a, b]$  be such that  $\gamma'(t_0) \neq 0$ . Writing  $\gamma(t)$  as

$$\gamma(t) = \gamma(t_0) + \gamma'(t_0)(t - t_0) + O(t - t_0)$$

The linear part of the RHS defines a line through  $\gamma(t_0)$  :

$$l : t \mapsto l(t) = \gamma(t_0) + \gamma'(t_0)(t - t_0)$$

This line  $l$  is called the *tangent* to the curve  $\gamma$  at  $\gamma(t_0)$ . (See the Fig. 1.11. on page 28.) If  $\gamma'(t_0) = 0$ , then the tangent at  $\gamma(t_0)$  is not defined. Because, if  $\gamma'(t_0) = 0$ , then curves can exhibit a number of different behaviors in the vicinity of  $\gamma(t_0)$ .

**Example 1.3.2.** Consider  $\gamma : t \mapsto t^3, t \in [-1, 1]$ . This is a Jordan arc of class  $\mathbf{C}^1$ ,. In this case  $\gamma'(0) = 0$  but  $\gamma(0) = 0$  is NOT a singular point ...

**Example 1.3.3.** Consider  $\gamma : t \mapsto t^2 + it^3, t \in [-1, 1]$ . This is a Jordan arc of class  $\mathbf{C}^1$  and  $\gamma'(0) = 0$ . In this case  $\gamma(0) = 0$  is a singular point ... See Fig. 1.12. on page 29.

**Example 1.3.4.** Consider  $\gamma$  defined on  $[0, 1]$  by  $\gamma(t) = t^3 e^{2\pi i/t}$  if  $t \in (0, 1]$  and  $\gamma(0) = 0$ . Note that  $\gamma'(0) = 0$  and  $\gamma$  is continuously differentiable on  $[0, 1]$ . Note that for any real  $x$  we have:

$$|\gamma(x)| = |x^3 e^{2\pi i/x}| = |x^3| |e^{2\pi i/x}| = |x^3|$$

Hence, for any  $s, t$  with  $0 \leq s < t \leq 1$  we have

$$|\gamma(s)| = s^3 < t^3 = |\gamma(t)|$$

Hence  $\gamma(t), t \in [0, 1]$  is a Jordan arc of class  $\mathbf{C}^1$ . If  $t \rightarrow 0$ ,  $\gamma(t)$  "spirals infinitely often" around 0 and it is clear that  $\gamma(0) = 0$  is a singular point ... See Fig. 1.13. on page 29.

If  $\gamma : t \mapsto \gamma(t), t \in [a, b]$  is a smooth Jordan arc, then by definition  $\gamma'(t) \neq 0$  for all  $t \in I$ , so that the tangent is defined at each point. A Jordan curve  $\gamma : I = [a, b] \rightarrow \mathbb{C}$  satisfying

1.  $\gamma \in \mathbf{C}^1$ .

2.  $\gamma'(t) \neq 0$  for all  $t$ .
3.  $\gamma'(a) = \gamma'(b)$

is called a **smooth Jordan curve**. If  $\gamma$  and  $\lambda$  are two smooth Jordan curves intersecting in  $z_0 = \gamma(t_0) = \lambda(s_0)$ , then the angle  $\theta$  between the tangent to  $\gamma$  at  $\gamma(t_0)$  and the tangent to  $\lambda$  at  $\lambda(s_0)$  is called the angle between  $\lambda$  and  $\gamma$ . If  $\theta = m\pi$  for some integer  $m$ , then  $\gamma$  and  $\lambda$  are said to *touch* at  $z_0$ . See Fig. 1.14. on page 30.

### 1.3.2 Integrals

Generally, if  $\phi(t) = u(t) + iv(t)$ ,  $t \in [a, b]$  is a continuous function of  $t$ , then the definite integral is defined by:

$$\int_a^b \phi(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt$$

We want to derive a useful formula:

$$\left| \int_a^b \phi(t) dt \right| \leq \int_a^b |\phi(t)| dt \quad (3)$$

*Proof.* We may assume that  $\int_a^b \phi(t) dt \neq 0$ . Let  $\theta$  be the argument of the complex number  $\int_a^b \phi(t) dt$ . Then

$$\left| \int_a^b \phi(t) dt \right| = e^{-i\theta} \int_a^b \phi(t) dt = \int_a^b e^{-i\theta} \phi(t) dt = \int_a^b \Re(e^{-i\theta} \phi(t)) dt + i \int_a^b \Im(e^{-i\theta} \phi(t)) dt$$

where the imaginary part is zero. Note that

$$\Re(e^{-i\theta} \phi(t)) \leq |e^{-i\theta} \phi(t)| = |\phi(t)|$$

Hence

$$\left| \int_a^b \phi(t) dt \right| = \int_a^b \Re(e^{-i\theta} \phi(t)) dt \leq \int_a^b |\phi(t)| dt$$

as desired. □

The chain rule is also true for the composition of a complex-valued function  $\gamma(t) \in \mathbf{C}^1$  of real variable  $t$  and a holomorphic function  $f(z)$  of a complex variable  $z$ .

**Theorem 1.3.5.** *Let  $f(z)$  be a holomorphic function defined on a region  $D$  and let  $\gamma(t)$  be a function in  $\mathbf{C}^1$  defined on an interval  $I$ . If  $\gamma(t) \in D$  for all  $t \in I$ , then the composite function  $f(\gamma(t))$  is also in  $\mathbf{C}^1$ .*

$$I \subseteq \mathbb{R} \xrightarrow{\gamma} \text{im } \gamma \subseteq D \xrightarrow{f} \text{im } f \subseteq \mathbb{C}$$

and the chain rule is valid

$$\frac{d}{dt} f(\gamma(t)) = f'(\gamma(t)) \cdot \gamma'(t)$$

Let  $\gamma$  be curve in  $\mathbb{C}$  and let  $D$  be a region in  $\mathbb{C}$ . If  $\gamma(t) \in D$  for all  $t \in I$ , we say that  $\gamma$  is in  $D$  and we write  $\gamma \subset D$ . Let  $f(z)$  be a continuous function on  $D$ . A holomorphic function  $F(z)$ , defined on  $D$  satisfying  $F'(z) = f(z)$ , is called a **primitive function** of  $f(z)$  on  $D$ . If, for example,  $f(z)$  is given by

$$f(z) = \sum_{n=0}^{\infty} a_n (z - c)^n$$

where the power series converges on  $U_r(c)$  for some  $r$  with  $0 < r \leq +\infty$ . Then, by Theorem 1.2.6

$$F(z) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (z - c)^{n+1}$$

is a primitive function of  $f(z)$  on  $U_r(c)$ .

If  $F(z)$  is a primitive function of  $f(z)$  on  $D$  and if  $\gamma$  is a curve of class  $\mathbf{C}^1$ , we have

$$\frac{d}{dt} F(\gamma(t)) = f(\gamma(t)) \cdot \gamma'(t)$$

Writing  $F = (U(t) + iV(t))$ , we have

$$\frac{d}{dt} U(t) + i \frac{d}{dt} V(t) = f(\gamma(t)) \cdot \gamma'(t)$$

Integrating both sides from  $a$  to  $b$ , we have

$$F(\gamma(b)) - F(\gamma(a)) = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

This formula suggests the following definition:

**Definition 1.3.6.** Let  $\gamma$  be a curve of class  $\mathbf{C}^1$  on  $D$  and let  $f$  be a continuous function of  $z$  on  $D$ . The integral

$$\int_a^b f(\gamma(t)) \gamma'(t) dt$$

is called the *integral of  $f(z)$  along  $\gamma$*  and denoted by

$$\int_{\gamma} f(z) dz$$

Thus

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

The curve  $\gamma$  is called the *path of integration* of this integral.

If a primitive function  $F(z)$  of  $f(z)$  exists on  $D$ , then by the above definition, we have

$$\int_{\gamma} f(z) dz = F(\beta) - F(\alpha)$$

where  $\alpha = \gamma(a)$  and  $\beta = \gamma(b)$ . Therefore, in this case, the value of  $\int_{\gamma} f(z)dz$  depends only on the endpoints, and is independent of the choice of curve connecting  $\alpha$  and  $\beta$  on  $D$ . In particular, if  $\gamma$  is a closed curve, then  $\int_{\gamma} f(z)dz = 0$ .

If  $f(z)$  is continuous on  $D$ , the following inequality:

$$\left| \int_{\gamma} f(z)dz \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt$$

is an obvious consequence of the (3).

An integral  $\int_{\gamma} f(z)dz$  along a curve  $\gamma$  of class  $\mathbf{C}^1$  can be obtained as a limit of finite sums

$$\sum_{k=1}^m f(\zeta_k)(z_k - z_{k-1})$$

The details of this procedure are as follows: We choose  $m + 1$  points  $t_0, \dots, t_m$  in  $[a, b]$  such that  $a = t_0 < t_1 < \dots < t_m = b$  and we put  $\Delta = \{t_0, \dots, t_m\}$ . The interval  $[a, b]$  is divided into  $m$  intervals  $[t_{k-1}, t_k]$ ,  $k = 1, \dots, m$ , by these  $m + 1$  points. We call  $\Delta$  a *partition* and we denote the maximal length of the intervals  $[t_{k-1}, t_k]$  by  $\delta[\Delta]$ :

$$\delta[\Delta] := \max_k \{t_{k-1} - t_k\}$$

For each  $k$  we pick  $\tau_k \in [t_{k-1}, t_k]$  and putting  $\zeta_k = \gamma(\tau_k)$  and  $z_k = \gamma(t_k)$  we find a finite sequence of points on  $\gamma$

$$z_0 = \gamma(a), \quad \zeta_1, z_1, \dots, \zeta_m, z_m = \gamma(b)$$

which are ordered according to increasing values of  $t$ . This sequence determines a finite sum:

$$\sigma_{\Delta} := \sum_{k=1}^m f(\zeta_k)(z_k - z_{k-1})$$

Now,

$$\int_{\gamma} f(z)dz = \lim_{\delta[\Delta] \rightarrow 0} \sum_{k=1}^m f(\zeta_k)(z_k - z_{k-1})$$

Since  $f(\gamma(t))$  is defined on the closed interval  $[a, b]$ , it is a uniformly continuous function of  $t$ . That is, for each  $\epsilon > 0$ , there exists a  $\delta(\epsilon)$  such that

$$|f(\gamma(t)) - f(\gamma(s))| < \epsilon \quad \text{whenever } |t - s| < \delta(\epsilon) \text{ and } s, t \in [a, b]$$

Since

$$\int_{\gamma} f(z)dz = \int_a^b f(\gamma(t)) \gamma'(t) dt = \sum_{k=1}^m \int_{t_{k-1}}^{t_k} f(\gamma(t)) \gamma'(t) dt$$

and since from

$$z_k - z_{k-1} = \gamma(t_k) - \gamma(t_{k-1}) = \int_{t_{k-1}}^{t_k} \gamma'(t) dt$$

we have

$$f(\zeta_k)(z_k - z_{k-1}) = \int_{t_{k-1}}^{t_k} f(\zeta_k)\gamma'(t)dt = \int_{t_{k-1}}^{t_k} f(\gamma(\tau_k))\gamma'(t)dt$$

Hence

$$\int_{\gamma} f(z)dz - \sum_{k=1}^m f(\zeta_k)(z_k - z_{k-1}) = \sum_{k=1}^m \int_{t_{k-1}}^{t_k} (f(\gamma(t)) - f(\gamma(\tau_k)))\gamma'(t)dt$$

we have If  $\Delta$  is a partition s.t.  $\delta[\Delta] < \delta(\epsilon)$ , then for all  $t \in [t_{k-1}, t_k]$ , we have  $|t - \tau_k| < \delta(\epsilon)$ , then for all  $t \in [t_{k-1}, t_k]$ , we have  $|t - \tau_k| < \delta(\epsilon)$ . Hence  $|f(\gamma(t)) - f(\gamma(\tau_k))| < \epsilon$ . Hence

$$\begin{aligned} \left| \int_{\gamma} f(z)dz - \sum_{k=1}^m f(\zeta_k)(z_k - z_{k-1}) \right| &= \left| \sum_{k=1}^m \int_{t_{k-1}}^{t_k} (f(\gamma(t)) - f(\gamma(\tau_k)))\gamma'(t)dt \right| \\ &\leq \sum_{k=1}^m \left| \int_{t_{k-1}}^{t_k} (f(\gamma(t)) - f(\gamma(\tau_k)))\gamma'(t)dt \right| \\ &\leq \sum_{k=1}^m \int_{t_{k-1}}^{t_k} |f(\gamma(t)) - f(\gamma(\tau_k))| |\gamma'(t)| dt \\ &\leq \epsilon \sum_{k=1}^m \int_{t_{k-1}}^{t_k} |\gamma'(t)| dt \\ &= \epsilon \int_a^b |\gamma'(t)| dt \end{aligned}$$

as desired.

### 1.3.3 Cauchy's integral formula for circles

Let  $f(z)$  be a holomorphic function on the region  $D \in \mathbb{C}$ , and  $U_r(c)$  a disk with center  $c$  and radius  $r \in (0, +\infty)$ , satisfying  $[U_r(c)] \subset D$ . Then, the following theorem holds:

**Theorem 1.3.7.** *Let  $\gamma : \theta \mapsto c + re^{i\theta}$ ,  $\theta \in [0, 2\pi]$  be the circle with center  $c$  and radius  $r$ . If  $w$  is a point of  $U_r(c)$ , then the value  $f(w)$  is given by*

$$f(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - w} dz \quad (4)$$

See Fig. 1.17. on page 35.

Formula (4) is a special case of Cauchy's integral formula, which we will prove later. Putting  $w = c$  we have

$$f(c) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - c} dz = \int_0^{2\pi} \frac{f(\gamma(\theta))}{\gamma(\theta) - c} \gamma'(\theta) d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(\gamma(\theta)) d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(c + re^{i\theta}) d\theta$$

The above formula shows that the value  $f(c)$  of a holomorphic function  $f(z)$  at a point  $c$  is equal to the average of the values  $f(c + re^{i\theta})$  taken on the points of a circle with center  $c$ . Therefore, the above formula is called the Mean Value Theorem.

Now we prove the integral formula (4). Let  $\gamma_\epsilon : \theta \mapsto w + \epsilon e^{i\theta}, \theta \in [0, 2\pi]$  represent a circle with center  $w$  and radius  $\epsilon$  satisfying  $\epsilon \in (0, r - |w - c|)$ . From

$$\frac{1}{2\pi i} \int_{\gamma_\epsilon} \frac{f(z)}{z - w} dz = \frac{1}{2\pi} \int_0^{2\pi} f(w + \epsilon e^{i\theta}) d\theta$$

we have

$$\lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int_{\gamma_\epsilon} \frac{f(z)}{z - w} dz = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi} \int_0^{2\pi} f(w + \epsilon e^{i\theta}) d\theta = f(w)$$

Therefore, in order to prove (4), it suffices to show

$$\frac{1}{2\pi i} \int_{\gamma_\epsilon} \frac{f(z)}{z - w} dz = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - w} dz$$

We first need a lemma:

**Lemma 1.3.8.** *Let  $f(z)$  be a holomorphic function on a region  $D$  and let  $\gamma_0 : t \mapsto \gamma_0(t)$  and  $\gamma_1 : t \mapsto \gamma_1(t), t \in [a, b]$  be closed curves of class  $\mathbf{C}^1$  in  $D$ . If for all  $t \in [a, b]$ , the segments  $L_t$  connecting  $\gamma_0(t)$  and  $\gamma_1(t)$ , represented by*

$$L_t = \{(1 - s)\gamma_0(t) + s\gamma_1(t) : s \in [0, 1]\}$$

*are in  $D$ . Then*

$$\int_{\gamma_0} f(z) dz = \int_{\gamma_1} f(z) dz$$

*Proof.* Let

$$\gamma_s(t) := (1 - s)\gamma_0(t) + s\gamma_1(t), \quad t \in [a, b]$$

for each  $s \in [0, 1]$ . See Fig.1.18. on page 38. It suffices to show that

$$\int_{\gamma_s} f(z) dz$$

is independent of  $s$ , and to prove this it suffices to show that

$$\frac{d}{ds} \int_{\gamma_s} f(z) dz = 0$$

Let  $\Gamma(t, s) := \gamma_s(t)$ . We see that  $\Gamma$  is a function of two (real) variables  $t, s$  defined on the rectangle  $K = \{(t, s) : t \in [a, b], s \in [0, 1]\}$  with continuous partial derivatives on  $K$  :

$$\begin{aligned} \Gamma_t &= (1 - s)\gamma'_0(t) + s\gamma'_1(t) \\ \Gamma_s &= \gamma_1(t) - \gamma_0(t) \\ \Gamma_{ts} &= \Gamma_{st} = \gamma'_1(t) - \gamma'_0(t) \end{aligned}$$

Since

$$\int_{\gamma_s} f(z) dz = \int_a^b f(\gamma_s(t)) \gamma'_s(t) dt = \int_a^b f(\Gamma) \Gamma_t dt,$$

it suffices to show that

$$\frac{d}{ds} \int_a^b f(\Gamma) \Gamma_t dt = 0$$

Note that

$$\frac{\partial}{\partial s} f(\Gamma) = f'(\Gamma) \Gamma_s$$

Hence

$$\frac{\partial}{\partial s} f(\Gamma) \Gamma_t = f'(\Gamma) \Gamma_s \Gamma_t + f(\Gamma) \Gamma_{ts} = \frac{\partial}{\partial t} f(\Gamma) \Gamma_s \quad (5)$$

Therefore

$$\begin{aligned} \frac{d}{ds} \int_a^b f(\Gamma) \Gamma_t dt &= \int_a^b \frac{\partial}{\partial s} f(\Gamma) \Gamma_t dt = \int_a^b \frac{\partial}{\partial t} f(\Gamma) \Gamma_s dt \\ &= f(\Gamma(b, s)) \Gamma_s(b, s) - f(\Gamma(a, s)) \Gamma_s(a, s) \\ &= 0 \end{aligned}$$

since  $\Gamma(b, s) = \Gamma(a, s)$  and  $\Gamma_s(b, s) = \Gamma_s(a, s)$ . So we are done.  $\square$

*Proof of Theorem 1.3.7.* This is a direct consequence of the above lemma.  $\square$

If the curve  $\gamma_0$  of Lemma 1.3.8 is defined by  $\gamma_0 = c$  (i.e., a point in the complex plane), then  $\gamma'_0(t) = 0$ . Hence  $\int_{\gamma_0} f(z) dz = 0$ . This gives the following corollary to Lemma 1.3.8.

**Corollary 1.3.9.** *Let  $f(z)$  be holomorphic on  $D$ ,  $\gamma : t \mapsto \gamma(t), t \in [a, b]$  a closed curve of class  $\mathbf{C}^1$  in  $D$  and  $c \in D$ . If all segments connecting  $c$  and  $\gamma(t), t \in [a, b]$  are in  $D$  then*

$$\int_{\gamma_0} f(z) dz = 0$$

In the proof of Lemma 1.3.8 we did not use the explicit formula for  $\Gamma(t, s)$ . We used the existence and continuity of the partial derivatives  $\Gamma_t(t, s), \Gamma_s(t, s)$  and  $\Gamma_{ts}(t, s) = \Gamma_{st}(t, s)$  and assertion  $\Gamma(b, s) = \Gamma(a, s)$ .

Therefore, let us assume that there exists a continuous map  $\Gamma : (t, s) \mapsto \Gamma(t, s)$  defined on the rectangle  $K$ , taking values in  $D$  such that ... things above happen.

Let  $\gamma_s(t) = \Gamma(t, s)$  for  $s \in [0, 1]$ .  $\gamma_s$  can be viewed as a curve of class  $\mathbf{C}^1$  in  $D$ . Since we did not assume that the equality  $\Gamma(b, s) = \Gamma(a, s)$  holds, the curves  $\gamma_s$  are not necessarily closed. Note that we have

$$\frac{d}{ds} \int_{\gamma_s} f(z) dz = f(\Gamma(b, s)) \Gamma_s(b, s) - f(\Gamma(a, s)) \Gamma_s(a, s) \quad (6)$$

Now define  ${}_t\gamma(s) := \Gamma(t, s)$  for all  $t \in [a, b]$ .  ${}_t\gamma : s \mapsto {}_t\gamma(s), s \in [0, 1]$  can be viewed as a curve of class  $\mathbf{C}^1$  in  $D$  for all  $t \in [a, b]$  and

$$\int_{{}_t\gamma} f(z) dz = \int_0^1 f(\Gamma(t, s)) \Gamma_s(t, s) ds$$

Integrating both sides of (6) we obtain:

$$\int_0^1 \frac{d}{ds} \int_{\gamma_s} f(z) dz ds = \int_0^1 \left( f(\Gamma(b, s)) \Gamma_s(b, s) - f(\Gamma(a, s)) \Gamma_s(a, s) \right) ds$$

Note that

$$\text{LHS} = \int_{\gamma_1} f(z) dz - \int_{\gamma_0} f(z) dz$$

and

$$\text{RHS} = \int_{b\gamma} f(z) dz - \int_{a\gamma} f(z) dz$$

so we have proved:

**Theorem 1.3.10** (Cauchy). *Let  $f(z)$  be a holomorphic function on  $D$  and let  $\Gamma : (t, s) \mapsto \Gamma(t, s)$  be a continuous map defined on the rectangle  $K$  taking values in  $D$ , such that the partial derivatives  $\Gamma_t(t, s)$ ,  $\Gamma_s(t, s)$  and  $\Gamma_{ts}(t, s) = \Gamma_{st}(t, s)$  exist and are continuous on  $K$ . Let*

$$\gamma_s : t \mapsto \gamma_s(t) = \Gamma(t, s), t \in [a, b] \quad \text{and} \quad {}_t\gamma : s \mapsto {}_t\gamma(s) = \Gamma(t, s), s \in [0, 1]$$

Then

$$\int_{\gamma_1} f(z) dz - \int_{\gamma_0} f(z) dz = \int_{b\gamma} f(z) dz - \int_{a\gamma} f(z) dz$$

*Remark* (by myself). Note that this result is quite intuitive. See Fig. 1.21 on page 42. We may have

$$\int_{(a\gamma) \rightarrow (\gamma_1) \rightarrow (-b\gamma) \rightarrow (-\gamma_0)} f(z) dz = 0$$

by the corollary above, where  $(a\gamma) \rightarrow (\gamma_1) \rightarrow (-b\gamma) \rightarrow (-\gamma_0)$  is a closed curve in  $D$  and ....

*Remark.* Theorem 1.3.10 is a special case of Cauchy's Theorem, which will be proved in Section ????. The vital part of the proof of Theorem 1.3.10 is equality (5):

$$\frac{\partial}{\partial s} f(\Gamma) \Gamma_t = \frac{\partial}{\partial t} f(\Gamma) \Gamma_s$$

Actually, this equality is a generalization of the Cauchy-Riemann equations (1). To see this, replace  $t, s$  by  $x, y$  and let  $\Gamma(x, y) := x + iy$ . Note that  $\Gamma_x = 1, \Gamma_y = i$ . Then, equality (5) becomes:

$$\frac{\partial}{\partial y} f(x + iy) = i \frac{\partial}{\partial x} f(x + iy)$$

Writing  $f(x + iy)$  as  $u(x, y) + iv(x, y)$  we have

$$u_y + iv_y = i(u_x + iv_x) = iu_x - v_x$$

Hence  $u_x = v_y$  and  $u_y = -v_x$ , as desired.

In fact, by integrating both sides of the generalized C-R equations over the rectangle  $K$ :

$$\iint_K \frac{\partial}{\partial s} f(\Gamma) \Gamma_t dt ds = \iint_K \frac{\partial}{\partial t} f(\Gamma) \Gamma_s dt ds$$

we have

$$\begin{aligned} \text{LHS} &= \int_0^1 \frac{\partial}{\partial s} \left( \int_a^b f(\Gamma) \Gamma_t dt \right) ds = \int_a^b f(\Gamma(t, 1)) \Gamma_t(t, 1) dt - \int_a^b f(\Gamma(t, 0)) \Gamma_t(t, 0) dt \\ &= \int_{\gamma_1} f(z) dz - \int_{\gamma_0} f(z) dz \end{aligned}$$

and

$$\begin{aligned} \text{RHS} &= \int_a^b \frac{\partial}{\partial t} \left( \int_0^1 f(\Gamma) \Gamma_s ds \right) dt = \int_0^1 f(\Gamma(b, s)) \Gamma_s(b, s) ds - \int_a^b f(\Gamma(a, s)) \Gamma_s(a, s) ds \\ &= \int_{b\gamma} f(z) dz - \int_{a\gamma} f(z) dz \end{aligned}$$

so we arrive at Cauchy's Theorem 1.3.10.

### 1.3.4 Power series expansions

**Theorem 1.3.11.** *Let  $\gamma$  be a curve of class  $\mathbf{C}^1$ .*

1. *If the sequence of functions  $\{f_n(z)\}$  converges uniformly to  $f(z)$  on  $\gamma$ , then*

$$\int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz$$

2. *If the series of functions  $\sum_{n=1}^{\infty} f_n(z)$  converges uniformly to  $f(z)$  on  $\gamma$ , then*

$$\int_{\gamma} f(z) dz = \sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz$$

Let  $D$  be a region in  $\mathbb{C}$  and let  $c \in D$ . We want to consider disks  $U_r(c)$  with center  $c$  and radius  $r$ . First, let us assume that  $D \neq \mathbb{C}$ . Then there exists a maximum value of  $r$  for which  $U_r(c) \subset D$ . That is, there exists an  $r(c)$  satisfying

$$U_{r(c)}(c) \subset D \quad \text{but} \quad [U_{r(c)}(c)] \not\subsetneq D$$

If  $f(z)$  is a holomorphic function on  $D$  and  $r < r(c)$ , then the Cauchy's formula (4) holds with regard to the circle

$$\gamma_r : \theta \mapsto c + re^{i\theta}, \quad 0 \leq \theta \leq 2\pi$$

That is, we have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad z \in U_r(c)$$

**Theorem 1.3.12.** Let  $f(z)$  be a holomorphic function on  $D$  and let  $c \in D$ . Then  $f(z)$  can be extended in a power series

$$f(z) = \sum_{n=0}^{\infty} a_n (z - c)^n$$

which converges on  $U_{r(c)}$ . The coefficients  $a_n$  are given by:

$$a_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{(\zeta - c)^{n+1}} d\zeta, \quad 0 < r < r(c)$$

*Proof.* We may assume without loss of generality that  $c = 0$ . Fix an arbitrary  $z \in U_{r(0)}(0)$  and choose  $r$  such that  $|z| < r < r(0)$  (i.e., The circle  $\gamma_r$  must contain  $z$ .) Then we have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Since  $\zeta$  is a point on the circle  $\gamma_r$ , we can write  $\zeta = \gamma_r(\theta) = re^{i\theta}$ . Hence  $|\zeta| = r > |z|$  and therefore  $|z/\zeta| < 1$ . So we can write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta} \cdot \frac{1}{1 - z/\zeta} = \frac{1}{\zeta} \cdot \sum_{n=0}^{\infty} \left(\frac{z}{\zeta}\right)^n = \sum_{n=0}^{\infty} \frac{z^n}{\zeta^{n+1}}$$

Hence

$$\frac{f(\zeta)}{\zeta - z} = \sum_{n=0}^{\infty} \frac{f(\zeta)z^n}{\zeta^{n+1}}$$

The continuous function  $|f(\gamma(\theta))|$  assumes a maximum on  $[0, 2\pi]$ . Denoting this maximum by  $M$  we have

$$\left| \frac{f(\zeta)z^n}{\zeta^{n+1}} \right| \leq M \frac{|z|^n}{r^{n+1}}, \quad \sum_{n=1}^{\infty} \frac{|z|^n}{r^{n+1}} = \frac{1}{r - |z|} < +\infty$$

Therefore, the series

$$\sum_{n=0}^{\infty} \frac{f(\zeta)z^n}{\zeta^{n+1}}$$

converges uniformly on the circle  $\gamma_r$  and thus, by Theorem 1.3.11(2), we have

$$\int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} \int_{\gamma_r} \frac{f(\zeta)z^n}{\zeta^{n+1}} d\zeta = \sum_{n=0}^{\infty} \left( \int_{\gamma_r} \frac{f(\zeta)}{\zeta^{n+1}} d\zeta \right) z^n$$

Let

$$a_n := \frac{1}{2\pi i} \left( \int_{\gamma_r} \frac{f(\zeta)}{\zeta^{n+1}} d\zeta \right)$$

we arrive at

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

□

**Corollary 1.3.13.** *A function  $f(z)$  defined on a region  $D$  is holomorphic if and only if for each  $c \in D$  there exists a neighborhood  $U_\epsilon(c) \subset D$  on which  $f(z)$  can be expanded in an absolutely convergent power series*

$$f(z) = \sum_{n=0}^{\infty} a_n(z-c)^n \quad (\epsilon \text{ depends on } c)$$

*If  $f(z)$  is holomorphic on  $D$ , then  $f(z)$  is arbitrarily often differentiable on  $D$  and the derivatives  $f'(z), f''(z), \dots$  are all holomorphic on  $D$ .*

## 1.4 Properties of holomorphic functions

### 1.4.1 $m^{\text{th}}$ -order derivatives

In this section we want to deduce the above fact directly from Cauchy's integral formula (4) and find integral formulas for the  $m^{\text{th}}$ -order derivatives as well.

**Lemma 1.4.1.** *Let  $\gamma(t)$  and  $\psi(t)$  be complex-valued continuous functions defined on  $I = [a, b]$ . For each natural number  $n$*

$$g(z) = \int_a^b \frac{\psi(t)}{(\gamma(t) - z)^n} dt$$

*is a holomorphic function of  $z$  defined on  $\mathbb{C} - \gamma(I)$  and its derivative is given by*

$$g'(z) = n \int_a^b \frac{\psi(t)}{(\gamma(t) - z)^{n+1}} dt$$

We omit the details of this sub-subsection (read page 47-49). But one thing we shall keep in mind: Cauchy's integral formula can DETECT whether a point is inside a closed curve or not. In the special case (where  $\gamma_r$  is a circle), we have:

1.  $z \in U_r(c)$  : we can read off  $f(z)$  or  $f^{(m)}(z)$  through the integral.
2.  $z \notin U_r(c)$  : the integral is zero.

### 1.4.2 Limits of sequences of holomorphic functions

**Theorem 1.4.2.** *Let  $\{f_n(z)\}$  be a sequence of holomorphic functions defined on  $D$ , **converging uniformly on all compact subsets of  $D$** . Then  $f(z) = \lim_{n \rightarrow \infty} f_n(z)$  is holomorphic on  $D$  and its  $m$ -th derivatives is given by*

$$f^{(m)}(z) = \lim_{n \rightarrow \infty} f_n^{(m)}(z)$$

*and the sequence  $\{f_n^{(m)}(z)\}$  converges uniformly to  $f^{(m)}(z)$  on all compact subsets of  $D$ .*

*Proof.* Let  $U_r(c)$  be an arbitrary disk s.t.  $[U_r(c)] \subset D$ . Since, by assumption,  $\{f_n(z)\}$  converges uniformly on  $[U_r(c)]$ ,  $f(z) = \lim_{n \rightarrow \infty} f_n(z)$  is a continuous function of  $z$  on  $[U_r(c)]$ . (Why?). See Theorem 1.2.2. Let  $\gamma_r$  be the circle with center  $c$  and radius  $r$ . Then, by the integral formula, we have

$$f_n(z) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f_n(\zeta)}{\zeta - z} d\zeta \quad z \in U_r(c)$$

Since  $\{f_n(\zeta)\}$  converges to  $f(\zeta)$  on  $[U_r(c)]$ ,  $\{f_n(\zeta)/(\zeta - z)\}$  converges uniformly to  $f(\zeta)/(\zeta - z)$  on  $\gamma_r$  for a fixed  $z \in U_r(c)$ . (Why?) Therefore, by Theorem 1.3.11 (1),

$$\lim_{n \rightarrow \infty} \int_{\gamma_r} \frac{f_n(\zeta)}{\zeta - z} d\zeta = \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta$$

Hence

$$f(z) = \lim_{n \rightarrow \infty} f_n(z) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad z \in U_r(c)$$

Therefore  $f(z)$  is holomorphic on  $U_r(c)$  and since  $U_r(c)$  is an arbitrary disk on  $D$ , we see that  $f(z)$  is holomorphic on  $D$ . ... (For the rest of the proof, see page 50.)  $\square$

### 1.4.3 The main value theorem and the maximum principle

**Theorem 1.4.3** (Main value theorem). *If  $f$  is a holomorphic function on  $D \ni c$  and  $[U_r(c)] \subset D$ . Then*

$$f(c) = \frac{1}{2\pi} \int_0^{2\pi} f(c + r^{i\theta}) d\theta$$

By taking the absolute value, we obtain:

$$|f(c)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(c + r^{i\theta})| d\theta$$

Next, let  $f(z)$  be a holomorphic function on  $D$ . We want to consider the real-valued function  $|f(z)|$  on  $D$ .

**Theorem 1.4.4** (Maximum principle). *Unless  $f(z)$  is a constant function on  $D$ ,  $|f(z)|$  assumes no maximum on  $D$ .*

*Proof.* Assume that  $|f(z)|$  assumes a maximum value  $M$ , at, say  $z = c$ . We have to show that  $f(z)$  is a constant function on  $D$ . Choose  $r$  s.t.  $[U_r(c)] \subset D$ . Then

$$\frac{1}{2\pi} \int_0^{2\pi} M - |f(c + r^{i\theta})| d\theta \leq M - |f(c)| = 0$$

Since  $g_r(\theta) := M - |f(c + r^{i\theta})|$  is a continuous function of  $\theta$  and  $g_r(\theta) \geq 0$ , we conclude that  $g_r(\theta) = 0$ . Note that this holds for every small enough  $r$ . Hence, for every  $\varepsilon > 0$  s.t.  $[U_\varepsilon(c)] \subset D$ , whenever  $z \neq c$ , we may write:

$$z = c + r^{i\theta}$$

for some  $r \in (0, \varepsilon)$ , and have  $|f(z)| = M$ . Of course, when  $z = c$ ,  $|f(z)| = M$  already. Hence for every  $z \in U_\varepsilon(c)$ , we have  $f(z) = M$ . **That is, whenever  $f$  reaches the maximum value at some point  $c \in D$ , there is a open neighborhood  $U(c)$  s.t.  $f$  reaches the maximum on the entire  $U(c)$ .** Now, let

$$V := \{z \in D : |f(z)| = M\}$$

Then, based on our argument, for every  $c \in V$ , there is a open neighborhood  $U(c) \subset V$ . Hence  $V$  is open. On the other hand,  $V = |f|^{-1}(\{M\})$ , which is also closed. Therefore, since  $V \neq \emptyset$ , we conclude that  $V = D$ . This shows that  $|f(z)|$  is a constant on  $D$ , we then show that  $f(z)$  is a constant on  $D$ . If  $M = 0$ , we are done. So we may assume that  $M \neq 0$ . Write  $f(z) = u(x, y) + iv(x, y)$  where  $z = x + iy$ . Then

$$\begin{aligned} uu_x + vv_x &= \frac{1}{2} \frac{\partial}{\partial x} (u^2 + v^2) = 0 \\ uu_y + vv_y &= \frac{1}{2} \frac{\partial}{\partial y} (u^2 + v^2) = 0 \end{aligned}$$

That is

$$\begin{pmatrix} u_x & v_x \\ u_y & v_y \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{0}$$

Since  $M > 0$ , we have  $|f(z)|^2 = u^2 + v^2 = M^2 > 0$ , so  $\begin{pmatrix} u \\ v \end{pmatrix} \neq \mathbf{0}$ . By Cauchy-Riemann, we have

$$0 = \det \begin{pmatrix} u_x & v_x \\ u_y & v_y \end{pmatrix} = u_x v_y - v_x u_y = u_x^2 + v_x^2 = v_y^2 + u_y^2$$

Hence, both  $u, v$  are constant, as desired.  $\square$

**Corollary 1.4.5.** *Let  $D$  be a bounded region and let  $f$  be defined and continuous on  $[D]$  and holomorphic on  $D$ . Then  $|f(z)|$  assumes a maximum at a point on  $[D] - D$ .*

*Proof.* We may assume that  $D$  is not a constant. Since  $D$  is bounded,  $[D]$  is also bounded. Together with the fact that  $[D]$  is closed, we conclude that  $[D]$  is compact by Heine-Borel. Since  $|f(z)|$  is continuous on  $[D]$ , by the extreme value theorem,  $|f(z)|$  assumes a maximum at a point  $c \in [D]$ . Since  $|f(z)|$  is holomorphic on  $D$ , by the maximum value principle,  $|f(z)|$  assumes no maximum on  $D$ . Hence  $c \in [D] - D$ .  $\square$

#### 1.4.4 Isolated singularities

Let  $D$  be a region and  $c \in D$ . In this section we consider functions that are holomorphic on  $D - \{c\}$ .

**Theorem 1.4.6.** *A function  $f(z)$  that is holomorphic on  $D - \{c\}$  can be expanded in a power series*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - c)^n$$

which converges absolutely on  $U_{r(c)}(c) - \{c\}$ . Its coefficients are given by

$$a_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{(\zeta - c)^{n+1}} d\zeta, \quad 0 < r < r(c)$$

*Remark.* The power series  $\sum_{n=-\infty}^{\infty} a_n(z - c)^n$  is said to converge absolutely on  $U_{r(c)}(c) - \{c\}$  if the series

$$\sum_{n=1}^{\infty} a_n(z - c)^n \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{a_{-n}}{(z - c)^n}$$

both converge absolutely for  $0 < |z - c| < r(c)$ . A power series of the form  $\sum_{n=-\infty}^{\infty} a_n(z - c)^n$  is called a **Laurent series**.

*Proof.* W.l.o.g. we may assume that  $c = 0$ . Fix a point  $w$  and choose  $\varepsilon$  and  $r$  s.t.  $0 < \varepsilon < |w| < r < r(0)$ . Let

$$g(z) = \frac{f(z) - f(w)}{z - w}$$

Then  $g$  is a holomorphic function on  $D - \{0, w\}$ . By Theorem 1.3.12,  $f(z)$  can be expanded in an absolutely converging power series in some neighborhood of  $w$  :

$$f(z) = f(w) + b_1(z - w) + b_2(z - w)^2 + \dots$$

Therefore

$$g(z) = \frac{f(z) - f(w)}{z - w} = b_1 + b_2(z - w) + b_3(z - w)^2 + \dots$$

Therefore,  $g(z)$  can be defined at  $z = w$  in such a way that  $g(z)$  becomes a holomorphic function on some neighborhood of  $w$  and therefore on the whole  $D - \{0\}$ . The  $w$  here is called *removable singularity*. Note that  $b_1 = f'(w)$ . Now, we can apply Lemma 1.3.8.

$$\int_{\gamma_\varepsilon} g(z) dz = \int_{\gamma_r} g(z) dz$$

(Because there is no singular points in between  $\gamma_\varepsilon$  and  $\gamma_r$  anymore.) That is,

$$\text{LHS} := \int_{\gamma_\varepsilon} \frac{f(z) - f(w)}{z - w} dz = \int_{\gamma_r} \frac{f(z) - f(w)}{z - w} dz =: \text{RHS}$$

Note that applying the integral formula (4) to  $h(z) \equiv 1$ , we have:

$$1 = h(w) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{h(z)}{z - w} dz = \frac{1}{2\pi i} \int_{\gamma_r} \frac{1}{z - w} dz$$

while by the corollary to Lemma 1.3.8:

$$\int_{\gamma_\varepsilon} \frac{1}{z - w} dz = 0$$

Note that

$$\text{LHS} = \int_{\gamma_\varepsilon} \frac{f(z) - f(w)}{z - w} dz = \int_{\gamma_\varepsilon} \frac{f(z)}{z - w} dz - f(w) \int_{\gamma_\varepsilon} \frac{1}{z - w} dz = \int_{\gamma_\varepsilon} \frac{f(z)}{z - w} dz$$

and

$$\text{RHS} = \int_{\gamma_r} \frac{f(z) - f(w)}{z - w} dz = \int_{\gamma_r} \frac{f(z)}{z - w} dz - f(w) \int_{\gamma_r} \frac{1}{z - w} dz = \int_{\gamma_r} \frac{f(z)}{z - w} dz - f(w) \cdot 2\pi i$$

Therefore

$$f(w) = \frac{1}{2\pi i} \left( \int_{\gamma_r} \frac{f(z)}{z - w} dz - \int_{\gamma_\varepsilon} \frac{f(z)}{z - w} dz \right)$$

Replacing  $z$  by  $\zeta$  and  $w$  by  $z$  yields

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{\gamma_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta$$

For the first term of the right-hand side we have

$$\frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} a_n z^n, \quad \text{where } a_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta^{n+1}} d\zeta$$

Since on  $\gamma_\varepsilon$ ,  $\zeta$  is a point, so  $|\zeta/z| = \varepsilon/|z| < 1$ . Hence

$$\frac{1}{\zeta - z} = -\frac{1}{z} \cdot \frac{1}{1 - \zeta/z} = -\frac{1}{z} \cdot \sum_{n=0}^{\infty} \left( \frac{\zeta}{z} \right)^n = -\sum_{n=0}^{\infty} \frac{\zeta^n}{z^{n+1}}$$

so

$$\frac{f(\zeta)}{\zeta - z} = -\sum_{n=0}^{\infty} \frac{f(\zeta)\zeta^n}{z^{n+1}}$$

with the power series on the right-hand side converging uniformly absolutely on  $\gamma_\varepsilon$ . Hence, by Theorem 1.3.11 (2),

$$\frac{1}{2\pi i} \int_{\gamma_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = -\sum_{n=0}^{\infty} \frac{a_{-n-1}}{z^{n+1}}, \quad \text{where } a_{-n-1} = \frac{1}{2\pi i} \int_{\gamma_\varepsilon} \zeta^n f(\zeta) d\zeta$$

The series on the right-hand side converges for  $|z| > \varepsilon$ . Hence

$$f(z) = \sum_{n=0}^{\infty} a_n z^n + \sum_{n=0}^{\infty} \frac{a_{-n-1}}{z^{n+1}} = \sum_{n=-\infty}^{\infty} a_n z^n$$

Since  $\zeta^n f(\zeta)$  is a holomorphic function of  $\zeta$  on  $D - \{0\}$  and since all segments connecting  $\gamma_\varepsilon(\theta)$  and  $\gamma_r(\theta)$  are in  $D - \{0\}$ , we conclude from Lemma 1.3.8

$$\int_{\gamma_\varepsilon} \zeta^n f(\zeta) d\zeta = \int_{\gamma_r} \zeta^n f(\zeta) d\zeta$$

Replacing  $-n - 1$  by  $n$  in the expression for  $a_{-n-1}$ , we see that we can write for all indices

$$a_n = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(\zeta)}{\zeta^{n+1}} d\zeta$$

It is clear from Lemma 1.3.8 that  $a_n$  is independent of the choice of the radius  $r$  for  $0 < r < r(0)$ , of the path of integration  $\gamma_r$ . This proves that for all  $z$  with  $0 < |z| < r(0)$ , the series  $\sum_{n=-\infty}^{\infty} a_n z^n$  converges and that

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

The power series  $\sum_{n=-\infty}^{\infty} a_n z^n$  is composed of two power series,  $\sum_0^{\infty} a_n z^n$ , which converges absolutely for  $|z| < r(0)$  and  $\sum_{n=1}^{\infty} a_{-n} z^n$ , which converges absolutely for  $|z| > 0$ . Therefore,  $\sum_{n=-\infty}^{\infty} a_n z^n$  converges absolutely for  $0 < |z| < r(0)$ .  $\square$

The Laurent series

$$\sum_{n=-\infty}^{\infty} a_n (z - c)^n$$

is called the **Laurent expansion** of  $f(z)$  about  $c$ . The point  $c$  is called an isolated singular point of  $f(z)$  and the part of the Laurent expansion with negative exponents:

$$\sum_{n=-\infty}^{-1} a_n (z - c)^n = \cdots + \frac{a_{-2}}{(z - c)^2} + \frac{a_{-1}}{z - c}$$

is called the **principal part** of  $f(z)$  at  $c$ .

We distinguish the following three cases: (We are actually classifying isolated singularities)

CASE 1. There is no principal part. In this case we have for  $0 < |z - c| < r(c)$ ,

$$f(z) = a_0 + a_1(z - c) + a_2(z - c)^2 + \cdots$$

By defining  $f(c) = a_0$ , we make  $f(z)$  into a function which is also holomorphic at  $c$ . In this case,  $c$  is called a *removable singularity* of  $f(z)$ . Usually, one removes removable singularities from the start making  $f(z)$  into a function which is holomorphic on the whole  $D$ .

Let  $f(z)$  be holomorphic on  $D$ . A point  $c \in D$  such that  $f(c) = 0$  is called a zero of  $f(z)$ . Consider the Taylor expansion of  $f(z)$  around the zero  $c$ :

$$f(z) = a_0 + a_1(z - c) + a_2(z - c)^2 + \cdots$$

Then  $a_0 = f(c) = 0$ , and it is possible that some of the other coefficients are also equal to 0. If all coefficients  $a_1, a_2, \dots$  are equal to 0, then  $f(z)$  is identically equal to 0 in a neighborhood of  $c$ . Let us assume that not all coefficients are equal to 0 and that  $a_m$  is the first one that is not equal to 0. Then

$$f(z) = (z - c)^m (a_m + a_{m+1}(z - c) + a_{m+2}(z - c)^2 \cdots)$$

In this case,  $m$  is called the *order*, or the *multiplicity of the zero*  $c$ . Writing

$$f(z) = (z - c)^m g(z)$$

Then,  $g(z)$  is holomorphic on  $U_r(c)$  and  $g(c) = a_m \neq 0$ . Therefore,  $g(z) \neq 0$  on a sufficiently small  $U_\varepsilon(c)$ . Hence

$$f(z) = (z - c)^m g(z) \neq 0 \quad \text{for } 0 < |z - c| < \varepsilon$$

Thus, we have proved that there are no other zeros in a sufficiently small neighborhood of a zero of a holomorphic function. The zero  $c$  is therefore "**isolated**".

CASE 2. The principal part has only a finite number of terms. That is, for  $0 < |z - c| < r(0)$ , we have

$$f(z) = \frac{a_{-m}}{(z - c)^m} + \cdots + \frac{a_{-1}}{z - c} + \sum_{n=0}^{\infty} a_n(z - c)^n, \quad a_{-m} \neq 0$$

In this case  $c$  is called a *pole* of order  $m$  of the function  $f(z)$ . The function  $g(z)$ , defined by

$$g(z) = (z - c)^m f(z) = a_{-m} + a_{-m+1}(z - c) + \cdots$$

is holomorphic on  $U_{r(c)}(c)$  and  $g(c) = a_{-m} \neq 0$ . Therefore,  $g(z) \neq 0$  on a sufficiently small neighborhood  $U_\varepsilon(c)$ . From

$$f(z) = \frac{1}{(z - c)^m} g(z)$$

we have

$$\lim_{z \rightarrow c} |f(z)| = \frac{|g(z)|}{|z - c|^m} = +\infty$$

In this case, we write  $f(c) = +\infty$ . We will return to the meaning of  $\infty$  later.

Putting  $h(z) = 1/g(z)$  we have

$$\frac{1}{f(z)} = (z - c)^m h(z)$$

and  $h(z)$  is holomorphic on  $U_\varepsilon(c)$  and  $h(z) \neq 0$  for  $z \in U_\varepsilon(c)$ . Therefore, the pole of the  $m^{\text{th}}$ -order  $c$  of  $f(z)$  is a zero of the  $m^{\text{th}}$ -order of  $1/f(z)$ . Conversely, it is also true that ...

CASE 3. The principal part is an infinite series. That is,  $a_{-n} \neq 0$  for infinitely many natural numbers  $n$ . In this case,  $c$  is called an *essential singularity* of  $f(z)$ . As the following theorem shows, the behavior of  $f(z)$  in the neighborhood of an essential singularity is very complicated.

**Theorem 1.4.7** (Weierstrass's Theorem). *Let  $c$  be an essential singularity of  $f(z)$  and  $w$  be a complex number. It is possible to find a sequence  $\{z_n\}$  of points converging to  $c$  such that  $\lim_{n \rightarrow \infty} f(z_n) = w$ . It is also possible to find a sequence  $\{z_n\}$  converging to  $c$  such that  $\lim_{n \rightarrow \infty} |f(z_n)| = +\infty$ . In other words, in an arbitrary small neighborhood of a essential singularity  $c$ , the image of  $f$  is dense in  $\mathbb{C}$ .*

We first prove a proposition:

**Proposition 1.4.8.** *If  $c$  is an isolated singularity of  $f(z)$  such that  $f$  is bounded on some neighborhood of  $c$ , then  $c$  is a removable singularity.*

*Proof.* By assumption there exists a  $\delta > 0$  and a  $M$  s.t.  $|f(z)| \leq M$  for all  $|z - c| < \delta$ . Recall that

$$a_{-n} = \frac{1}{2\pi i} \int_{\gamma_r} (\zeta - c)^{n-1} f(\zeta) d\zeta$$

Hence

$$|a_{-n}| \leq \frac{1}{2\pi} 2\pi r M = rM$$

This holds for arbitrarily small  $r$ , forcing  $a_{-n}$  to be zero. □

*Proof of Theorem 1.4.7.* We first show that  $f(z)$  can go to  $+\infty$ . Suppose the contrary, that is, if there does not exist a sequence  $\{z_n\}$  converging to  $c$  s.t.  $f(z_n) \rightarrow +\infty$ , then  $f$  is bounded in a small neighborhood of  $c$ . By our above proposition,  $c$  is removable, contradicting our assumption that  $c$  is essential.

Then we show that  $f(z_n)$  may converge to an arbitrary number with a proper choice of  $\{z_n\}$ . Fix an arbitrary  $w \in \mathbb{C}$ . Suppose the contrary that  $\nexists$  sequence  $\{z_n\}$  converging to  $c$  s.t.  $f(z_n) \rightarrow w$ . That means  $f(z)$  would eventually "keep a distance" from  $w$  no matter which way you try to approach. That is,  $\exists \varepsilon > 0, \delta > 0$  s.t.

$$|f(z) - w| \geq \varepsilon \quad \text{whenever } 0 < |z - c| < \delta$$

Now, define

$$g(z) := \frac{1}{f(z) - w}$$

Note that  $g$  is holomorphic on  $0 < |z - c| < \delta$  and bounded by  $1/\varepsilon$ . Hence, by our proposition,  $c$  is a removable singularity of  $g$ . So we can define  $g(c)$  so that  $g$  is holomorphic on the entire  $|z - c| < \delta$ . Now we have 2 cases:

1.  $g(c) = 0$ . Then suppose that  $c$  is the  $m^{\text{th}}$ -order zero of  $g$ , i.e.,  $g(z) = (z - c)^m h(z)$ . Then

$$f(z) = w + \frac{1}{(z - c)^m h(z)}$$

so  $c$  is a  $m^{\text{th}}$ -order pole of  $f$ . Contradiction.

2.  $g(c) \neq 0$ . Then  $1/g$  is holomorphic around  $c$ . Now we have

$$f(z) = w + \frac{1}{g(z)}$$

Meaning that  $c$  is also a removable singularity of  $f$ . Contradiction. □

### 1.4.5 Entire functions

A function holomorphic on the whole complex plane  $\mathbb{C}$  is called an *entire function*. If  $f(z)$  is an entire function, then, by Theorem 1.3.12,  $f(z)$  can be expanded in a power series

$$f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

which converges absolutely on the whole complex plane  $\mathbb{C}$ . We distinguish the following three cases:

1.  $f(z) = a_0$
2.  $f(z) = a_0 + a_1z + \cdots + a_nz^n$
3.  $f(z) = a_0 + a_1z + \cdots$

The first case is trivial. For the second case, it is easy to see that  $|f(z)| \rightarrow +\infty$  as  $|z| \rightarrow \infty$ .

In the third case,  $f(z)$  is called a *transcendental entire function*. If  $f(z)$  is a transcendental entire function and  $w$  an arbitrary complex number, then there exists a sequence  $\{z_n\}$  s.t.  $\lim_{n \rightarrow \infty} |z_n| = \infty$  and  $\lim_{n \rightarrow \infty} f(z_n) = w$ ; There also exists a sequence  $\{z_n\}$  s.t.  $\lim_{n \rightarrow \infty} f(z_n) = \infty$  as  $|z_n| \rightarrow \infty$ .

To see this, let  $g(\zeta) = f(1/\zeta)$ . Then

$$g(\zeta) = \cdots + \frac{a_2}{\zeta^2} + \frac{z_1}{\zeta} + a_0$$

is a holomorphic function of  $\zeta$  on  $\mathbb{C} - \{0\}$ . Moreover, 0 is an essential singularity of  $g(\zeta)$ . So we can apply Theorem 1.4.7 to  $g(\zeta)$ .

**Theorem 1.4.9** (Liouville's Theorem). *A bounded entire function is a constant function.*

*Proof.* If  $f$  is a bounded entire function, then case 2 and 3 cannot happen. That is,  $f(z) = a_0$ .  $\square$

**Corollary 1.4.10** (The fundamental theorem of algebra).  $\overline{\mathbb{C}} = \mathbb{C}$ .

*Proof.* Let  $f(z) = a_0 + a_1z + \cdots + a_mz^m$  be a non-constant polynomial. Suppose the contrary that  $f$  has no zeros in  $\mathbb{C}$ . Then  $1/f(z)$  is an entire function. Moreover, it is bounded. **Why?** Fix a  $R > 0$ . For  $|z| > R$ ,  $1/f(z)$  is clearly bounded. For  $|z| \leq R$ , since  $1/f$  is continuous, by Heine-Borel,  $1/f(z)$  is bounded. Hence by Liouville,  $f$  is a constant. A contradiction.  $\square$

## References

- [1] Kunihiko Kodaira. *Complex Analysis*. Cambridge University Press, Cambridge, UK, 2007.